

Artificial Intelligence in Asset Pricing: A Critical Literature Review of Market Efficiency, Risk Premia, and Alpha Generation

*Critical Literature Review of the Research Problem
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1. Introduction

The application of artificial intelligence (AI) and machine learning (ML) to asset pricing has emerged as one of the most active research frontiers in empirical finance over the past decade. Traditional asset pricing rests on a small number of linear factor models – the Capital Asset Pricing Model, the Fama-French three- and five-factor models, and their many extensions – that impose strong a priori structure on the functional relationship between firm characteristics and expected returns. The proliferation of computing power, alternative data sources, and flexible non-parametric estimators has motivated a growing body of work that asks whether AI methods can describe, predict, and explain the cross-section and time series of asset returns more accurately than these canonical models, and whether such improvements are indicative of exploitable inefficiency or simply a better-fitting description of compensation for risk.

This review critically examines the research problem motivating the underlying thesis, *Artificial Intelligence in Asset Pricing: Evidence on Market Efficiency, Risk Premia and Alpha Generation*, by synthesising the empirical and theoretical literature along its three constituent strands. Section 2 outlines the theoretical background against which AI-based findings must be interpreted, namely the Efficient Market Hypothesis (EMH) and its adaptive reformulation. Section 3 reviews the methodological toolkit of AI in empirical asset pricing. Section 4 discusses evidence on market efficiency. Section 5 reviews the risk-premia measurement literature. Section 6 turns to alpha generation, including text- and sentiment-based strategies and reinforcement learning. Section 7 presents a comparative synthesis table of the AI method families surveyed. Section 8 offers a critical synthesis of methodological limitations – overfitting, the factor zoo, interpretability, and implementation frictions – that any thesis in this area must confront. Section 9 identifies the resulting research gaps, and Section 10 concludes.

2. Theoretical Background: Market Efficiency and Its Discontents

The Efficient Market Hypothesis, in its classical formulation, holds that security prices fully and instantaneously reflect all available information, such that consistent risk-adjusted outperformance should not be achievable. The hypothesis has historically been defended and challenged in roughly equal measure: decades of anomaly research documented in the cross-sectional asset pricing literature – value, size, momentum, low-volatility, and hundreds of further characteristics – can be read either as evidence of mispricing or as compensation for unmodelled sources of risk. The Adaptive Markets Hypothesis was proposed as a reconciling framework, arguing that market efficiency is not a fixed property but an evolutionary one, varying over time as the composition of market participants, their competitive pressures, and the information environment change. Empirical work applying rolling-window and machine-learning-based time-varying efficiency measures has lent support to this evolutionary view, finding that measures of return predictability cluster in identifiable regimes rather than converging monotonically to zero.

AI is theoretically double-edged with respect to this debate. On one hand, AI-enabled traders and algorithms can identify and arbitrage away pricing anomalies faster than human traders, which should push markets toward greater efficiency and accelerate the decay of previously documented

anomalies following their publication. On the other hand, the same technologies introduce new sources of correlated behaviour – model herding, feedback loops between competing algorithms, and latency-driven order-flow effects – that can generate fresh, temporary inefficiencies or amplify volatility, particularly during periods of stress. This tension between efficiency-enhancing and efficiency-disrupting effects of AI is a recurring theme that any evaluation of AI-based alpha claims must take seriously, since apparent predictability may reflect a transient artefact of the technology's own adoption rather than a stable structural feature of the market.

3. The Methodological Toolkit of AI in Empirical Asset Pricing

The methods literature has moved through roughly three overlapping generations. The first generation applies regularised linear and generalised linear models – ridge regression, the lasso, elastic net, and partial least squares – to shrink and select among the large sets of candidate predictors that characterise firm-level return prediction. The second generation introduces tree-based ensembles (random forests, gradient boosting) and feed-forward neural networks capable of capturing non-linear interactions among characteristics that linear models miss by construction. The landmark comparative study in this vein systematically benchmarks generalised linear models, dimension-reduction methods, boosted trees, random forests, and neural networks on the cross-section of US equity returns, and finds that tree-based methods and neural networks dominate, with the gains traced specifically to their ability to model non-linear interactions between characteristics such as momentum, liquidity, and volatility (Gu, Kelly, & Xiu, 2020). A third generation imposes economic structure directly on the learning problem rather than treating prediction as a purely statistical exercise. Kelly, Pruitt, and Su (2019) recast firm characteristics as proxies for dynamic factor loadings within an instrumented principal-components framework, while Chen, Pelger, and Zhu (2023) embed a no-arbitrage constraint inside a deep neural network to estimate the stochastic discount factor that must price all assets simultaneously, reporting that the constraint materially improves the model's ability to explain individual stock returns relative to an unconstrained prediction approach. Related work uses decision trees to construct parsimonious sets of basis assets (Bryzgalova, Pelger, & Zhu, 2020) and adaptive group-lasso methods to estimate the risk premia of characteristics non-parametrically (Freyberger, Neuhierl, & Weber, 2020).

A parallel body of work extends these methods beyond equities: Bianchi, Büchner, and Tamoni (2019) apply the machine-learning comparison of Gu, Kelly, and Xiu (2020) to the term structure of government bond returns, while autoencoder-based non-linear factor models generalise linear conditional factor pricing to allow factor loadings themselves to be learned functions of firm characteristics. Kelly and Xiu (2023) synthesise this emerging methods literature and emphasise that asset pricing poses distinctive challenges for machine learning relative to other prediction domains: an unusually low signal-to-noise ratio, an unknown and likely complex functional form linking predictors to returns, and information sets that are both very large and highly correlated across candidate predictors.

4. Evidence on Market Efficiency and Algorithmic Trading

Empirical assessment of how AI affects market efficiency has proceeded along two lines: microstructure studies of algorithmic and high-frequency trading, and macro-level studies of anomaly persistence. On the microstructure side, algorithmic trading is documented to have evolved from rule-based execution systems, designed principally to reduce transaction costs and market impact, toward predictive systems that learn non-linear relationships from limit-order-book dynamics and intraday volatility patterns (Kearns & Nevmyvaka, 2013). Recent theoretical treatments argue that AI-enabled trading narrows certain classical inefficiencies while simultaneously introducing new and potentially transient distortions that arise from competition and adaptation among similarly specified learning models, a finding that speaks directly to the AMH's evolutionary framing rather than a simple monotonic improvement in efficiency. Consistent with this, high-frequency intraday studies using order-book data find that the weak-form efficiency of exchange-traded funds fluctuates considerably even within a single trading day, alternating between clusters of high and low efficiency in a pattern the authors interpret as more consistent with an adaptive than a strictly efficient market. Complementary work using long-horizon deep-learning-based structural-change detection similarly concludes that the S&P 500's return-generating process is better described as evolving through regimes than as governed by a single, time-invariant equilibrium (Urquhart & McGroarty, 2016 provide earlier related non-ML evidence in the same tradition).

At the same time, commentary from policy institutions cautions that the diffusion of generative and large-language-model-based AI into trading and portfolio rebalancing, while still limited relative to established quantitative strategies, could raise trading volumes and correlated behaviour across funds using similar architectures, potentially increasing episodic volatility even as average pricing efficiency improves. This dual efficiency-volatility narrative is echoed by empirical studies of AI-driven algorithmic trading across global exchanges, which report simultaneous improvements in price discovery and bid-ask spreads alongside heightened volatility during periods of macroeconomic uncertainty. Taken together, the market-efficiency literature does not support a simple verdict that AI has made markets uniformly more or less efficient; rather, it supports the more nuanced claim that AI accelerates the adaptive dynamics that the AMH describes, compressing some anomalies while periodically generating others.

5. Risk Premia Measurement: Evidence and Open Questions

The risk-premia strand of the literature is the most methodologically mature. Gu, Kelly, and Xiu (2020) report that machine learning forecasts roughly double the out-of-sample explanatory power (R^2) of leading regression-based benchmarks and translate into materially larger Sharpe ratios for investors who condition portfolio construction on machine-learning forecasts rather than traditional linear signals. Because all methods examined converge on a similar set of dominant predictive characteristics – variants of momentum, liquidity, and volatility – the authors interpret the gains from flexible methods as arising primarily from better modelling of interactions among a fairly stable set of underlying signals rather than from the discovery of wholly new return drivers. Chen, Pelger, and Zhu (2023) extend this logic by showing that once no-arbitrage restrictions are imposed, most individual characteristics enter the stochastic discount factor in an approximately linear fashion when considered in isolation, and that the incremental value of deep non-linear

architectures comes specifically from capturing interaction effects across several characteristics simultaneously – a finding that helps explain why linear factor models have performed reasonably well historically despite their restrictive functional form.

A separate concern within this strand is whether machine-learning risk-premia estimates are robust to trading frictions. Avramov, Cheng, and Metzker (2020) raise the concern that machine-learning portfolios constructed without regard to transaction costs and short-sale constraints may look far more profitable in backtests than they would be in implementable practice, since predictive signal strength is often concentrated in small, illiquid, and high-turnover securities. Responding to this critique, subsequent work incorporates trading-friction constraints directly into the portfolio optimisation problem, seeking machine-learning portfolios that remain attractive net of realistic implementation costs (Bryzgalova, Pelger, & Zhu, 2020). This friction-aware line of research is important because it reframes the debate from whether AI can forecast returns statistically to whether AI-based risk-premia estimates constitute economically exploitable, implementable strategies – a distinction with direct bearing on any claim of market inefficiency.

6. Alpha Generation: Text, Sentiment, and Reinforcement Learning

A third strand of the literature moves beyond structured firm characteristics to unstructured data and sequential decision-making. Natural language processing methods extract sentiment signals from news articles, earnings-call transcripts, and social media, building on early evidence that media pessimism exerts measurable downward pressure on prices and that this relationship can be systematically quantified (Tetlock, 2007). More recent studies combine lexicon-based and transformer-based sentiment scores with financial news and microblog data to forecast next-period returns and volatility, generally reporting that sentiment-augmented models improve on a passive benchmark, though the size and stability of the resulting alpha varies considerably with the asset universe, sample period, and sentiment-extraction methodology (Deveikyte, Geman, Piccari, & Provetti, 2022). A newer wave of work uses large language models not merely to score sentiment but to generate candidate predictive formulas ("formulaic alphas") directly, feeding the resulting features into downstream sequence models; early results suggest such LLM-generated signals can improve price-level forecasts more reliably than return-level forecasts, underscoring that return prediction remains a harder and noisier problem than price-level extrapolation.

Reinforcement learning constitutes a second alpha-generation paradigm, framing portfolio and trade execution decisions as sequential control problems rather than one-shot prediction problems. Deep reinforcement learning agents have been applied to portfolio management, order execution, and multi-agent trading environments, with reported performance gains over static or equally weighted benchmarks in several out-of-sample settings (Guan & Liu, 2021). However, this literature is unusually self-critical about interpretability: because deep reinforcement learning policies are opaque by construction, a substantial and rapidly growing sub-literature is devoted specifically to explaining, rather than merely improving, agent behaviour, using tools such as attention mechanisms, SHAP values, and post-hoc feature-importance analysis to establish whether a trained agent's actions are consistent with a defensible investment rationale before

such strategies could plausibly be deployed or regulated. The prominence of this explainability agenda is itself an implicit acknowledgement within the field that predictive or backtested outperformance alone is an insufficient basis for claiming genuine, exploitable alpha.

7. Comparative Synthesis of AI Method Families

Table 1 draws together the method families reviewed in Sections 3–6, summarising the core idea behind each, representative studies, and the strengths and limitations reported in the literature. The table is intended to make explicit a pattern that recurs throughout this review: each successive generation of methods buys additional predictive flexibility at the cost of interpretability, robustness to trading frictions, or both – a trade-off that motivates the critical discussion in Section 8.

Method family	Representative studies	Core idea	Reported strengths	Key limitations
Regularised linear / GLM (ridge, lasso, elastic net, PLS)	Freyberger, Neuhierl & Weber (2020); Gu, Kelly & Xiu (2020)	Shrink and select among many correlated firm characteristics	Interpretable; computationally cheap; robust benchmark	Cannot capture non-linear interactions; dominated by trees/NNs in horse races
Tree ensembles & neural networks	Gu, Kelly & Xiu (2020)	Learn non-linear interactions among characteristics (momentum, liquidity, volatility)	Largest gains in out-of-sample R^2 and Sharpe ratio across studies	Highest overfitting risk; limited interpretability ("black box")
Structural / no-arbitrage-constrained models	Kelly, Pruitt & Su (2019); Chen, Pelger & Zhu (2023)	Embed economic theory (no-arbitrage, factor structure) directly into estimation	Ties predictions to an economically interpretable SDF; improves out-of-sample pricing	Computationally intensive; still depends on flexible functional forms for loadings
NLP / sentiment-based signals	Tetlock (2007); Deveikyte et al. (2022)	Extract predictive information from unstructured text and social media	Accesses information unavailable to structured-data models	Signal instability across periods; sensitive to extraction methodology
Reinforcement learning (portfolio / execution)	Guan & Liu (2021)	Model sequential trading decisions rather than one-shot return prediction	Directly optimises portfolio-level objectives; adapts to changing states	Severe interpretability challenges; explainability tooling still nascent

Table 1. Comparative summary of AI method families surveyed in this review.

8. Critical Synthesis: Methodological Limitations

8.1 Overfitting, Data Mining, and the Factor Zoo

The single most consequential critique facing this entire research programme predates the machine learning literature but applies to it with particular force. Harvey, Liu, and Zhu (2016) show that hundreds of published return-predicting factors have emerged from a research process characterised by extensive, largely unreported multiple testing, and argue that conventional statistical significance thresholds are far too permissive once this data-mining process is accounted for; they propose substantially higher hurdles for a newly discovered factor to be considered credible. Harvey and Liu's subsequent census of this so-called "factor zoo" documents that the rate

of new factor discovery has continued largely unabated despite these warnings. McLean and Pontiff (2016) provide complementary evidence that published anomalies experience significant performance decay after publication, consistent with a combination of statistical overfitting and genuine arbitrage-driven correction. Because machine learning models search an even larger and more flexible hypothesis space than the linear factor models these critiques were originally aimed at, the overfitting concern applies with at least equal, and arguably greater, force to AI-based return prediction: high in-sample R^2 or backtested Sharpe ratios are necessary but decidedly not sufficient evidence of a genuine, stable pricing relationship (Bailey & López de Prado, 2014).

8.2 Interpretability and the Black-Box Problem

A second, more distinctly AI-specific limitation concerns interpretability. Tree ensembles and deep neural networks are frequently described in the reviewed literature as "black-box" methods, in that the specific economic mechanism responsible for a given prediction is not directly recoverable from the model itself. This matters for asset pricing in a way that it may not for other prediction tasks, because a central purpose of asset pricing research is not merely to forecast returns but to understand the economic sources of risk compensation; a model that predicts well without articulating why is of limited use for testing or refining asset pricing theory, and offers little basis for assessing whether documented predictability reflects risk premia, behavioural mispricing, or statistical artefact. This concern has motivated a substantial explainable-AI sub-literature applying post-hoc attribution techniques, but these tools themselves face open questions about stability and faithfulness, and their use in finance remains comparatively immature relative to the predictive methods they are meant to explain.

8.3 Implementation, Transaction Costs, and Structural Breaks

Third, as discussed in Section 5, predictive accuracy measured on paper does not automatically translate into an implementable trading strategy once realistic transaction costs, capacity constraints, and short-selling frictions are considered. Related to this is the risk that AI models trained on historical data are poorly equipped to anticipate structural breaks – regulatory changes, shifts in market participant composition, or genuinely novel macroeconomic regimes – precisely the kind of non-stationarity that the Adaptive Markets Hypothesis identifies as central to how real markets evolve. This creates a tension at the heart of the research problem: the very adaptivity that motivates using flexible AI methods in the first place is also what limits confidence that a model estimated on one regime will continue to generate reliable risk-premia estimates or alpha in a different one.

9. Identified Research Gaps

Drawing the three strands together, several gaps in the existing literature can be identified that motivate further inquiry. First, most risk-premia studies to date document average, unconditional performance of machine-learning forecasts over long samples; comparatively little work directly tests whether the degree of achievable outperformance itself varies systematically with proxies for time-varying market efficiency, which would provide a more direct empirical bridge between the market-efficiency and risk-premia strands of the literature than currently exists. Second, the interpretability techniques developed largely within the reinforcement-learning

and explainable-AI literatures have been applied only sparingly to the canonical supervised-learning risk-premia models that dominate the empirical asset pricing literature, leaving open the question of which specific economic channels – momentum, liquidity, volatility, or their higher-order interactions – actually drive the superior performance of flexible methods across different markets and time periods. Third, the friction- and implementation-aware literature remains comparatively thin outside the largest and most liquid US equity markets, so the extent to which machine-learning-based alpha survives realistic implementation constraints in other asset classes or markets is not yet well established. Finally, relatively few studies integrate text-based and sentiment-based alpha signals with the structural, no-arbitrage-constrained risk-premia models reviewed in Section 3, despite the conceptual appeal of combining unstructured information with theoretically disciplined estimation. These gaps collectively motivate a research design capable of jointly evaluating market efficiency, risk-premia measurement, and alpha generation within a single, internally consistent empirical framework, rather than treating the three questions in isolation as the existing literature predominantly does.

10. Conclusion

The literature reviewed here demonstrates that AI methods have materially advanced the empirical toolkit of asset pricing, improving out-of-sample return prediction, offering new ways to estimate the stochastic discount factor under no-arbitrage restrictions, and opening novel sources of predictive signal in unstructured text and sequential trading behaviour. At the same time, this progress has been accompanied by a parallel and equally active critical literature warning that flexible, high-capacity models are especially prone to overfitting, that their opacity complicates economic interpretation, and that reported outperformance frequently does not survive realistic trading frictions or out-of-sample regime shifts. A defensible research design in this area must therefore treat predictive performance as necessary but not sufficient evidence of either genuine market inefficiency or robust alpha, and should explicitly incorporate economic structure, interpretability diagnostics, and implementation-cost considerations rather than relying on raw predictive accuracy alone. It is against this critical backdrop that the present thesis situates its own empirical investigation of AI's role in market efficiency, risk-premia measurement, and alpha generation.

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